

SIMATS ENGINEERING

ASSESSMENT TOOL 2 - Medium (LABORATORY PERFORMANCE)

COURSE CODE: CSA0921

**COURSE NAME: Programming in
Java**

COURSE OUTCOME COVERED

CO4: *Design and develop GUI-based user interfaces using Abstract Window Toolkit, Swing, and Applets by implementing event handling, layout managers, AWT controls, and menus. (BL3,BL5)*

Assessment Tool Weightage: 10%

QUESTIONS: 1

Total Marks: 50

ANNEXURE A

LABORATORY PERFORMANCE

Code of Conduct:

I _SIVARANGJENII B(192521375)_____ (Name / Reg No) *certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

SIVARANGJENII B

ANNEXURE

LABORATORY PERFORMANCE

1. Student Registration Applet

Develop a Java Applet for a university student registration form. The applet should display labels and input controls for Student Name, Register Number, Department, Gender, and Email ID. The application should use appropriate **AWT controls** such as Label, TextField, Checkbox, CheckboxGroup, and Button. Use a suitable **layout manager** to organize the controls. When the student clicks the **Submit** button, the entered information should be displayed within the applet window.

The student must demonstrate:

1. Applet initialization and architecture.
2. AWT controls.
3. Layout management.
4. Button event handling.
5. Appropriate event listener interface.
6. Display of user-entered information.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding & Program Design	20%	Clearly understands the problem and develops an effective design	Good understanding and logical design	Basic understanding with some gaps	Poor understanding and unclear design
Java Implementation & Concept Application	30%	Applies Java concepts correctly and effectively	Applies concepts with minor errors	Limited application of concepts	Incorrect or incomplete implementation
Program Execution & Output	20%	Program executes correctly and produces	Executes with minor errors	Partially executes with some errors	Fails to execute correctly

		accurate output			
Testing & Debugging	15%	Thoroughly tests and resolves errors independently	Tests and resolves most errors	Performs limited testing and debugging	Unable to test or debug effectively
Documentation & Presentation	15%	Complete, clear, and well-organized documentation	Good documentation with minor gaps	Basic documentation with some gaps	Incomplete or unclear documentation

CODE:

```
import java.applet.Applet;
import java.awt.*;
import java.awt.event.*;
```

```
public class StudentRegisterApplet extends Applet implements ActionListener
{
    Label title,nameLabel,regLabel,deptLabel,genderLabel,emailLabel,result;
    TextField name,reg,email;
    Choice dept;
    Checkbox male,female;
    CheckboxGroup gender;
    Button submit;

    public void init()
    {
        setLayout(new GridLayout(8,2,5,5));

        title=new Label("STUDENT REGISTRATION FORM");
        nameLabel=new Label("Student Name:");
        regLabel=new Label("Register Number:");
        deptLabel=new Label("Department:");
        genderLabel=new Label("Gender:");
        emailLabel=new Label("Email:");

        name=new TextField();
```

```
reg=new TextField();
email=new TextField();

dept=new Choice();
dept.add("IT");
dept.add("CSE");
dept.add("ECE");
dept.add("EEE");
dept.add("MECH");

gender=new CheckboxGroup();
male=new Checkbox("Male",gender,false);
female=new Checkbox("Female",gender,false);

submit=new Button("Submit");
result=new Label("");

add(title);
add(new Label(""));

add(nameLabel);
add(name);

add(regLabel);
add(reg);

add(deptLabel);
add(dept);

add(genderLabel);
Panel p=new Panel();
p.add(male);
p.add(female);
add(p);

add(emailLabel);
add(email);

add(submit);
add(result);

submit.addActionListener(this);
}
```

```

public void actionPerformed(ActionEvent e)
{
    String g=gender.getSelectedCheckbox()!=null?
    gender.getSelectedCheckbox().getLabel():"Not Selected";

    result.setText("Registered: "+name.getText()+" | "+reg.getText()+"
    " | "+dept.getSelectedItem()+" | "+g+" | "+email.getText());
}
}

```

OUTPUT:

Applet Viewer: StudentRegisterApplet.class
Applet

STUDENT REGISTRATION FORM

Student Name:

Register Number:

Department:

Gender: ☐ Male ☒ Female

Email:

Registered: SIVARANJENI B | 192521375 | IT | Female | siva@gmail.com

Applet started.